

The behavioral test you pick decides the brain anatomy you can see, and what matters is whether you measure errors or severity

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INTRODUCTION

Relating language deficits to brain damage requires two choices: which behavioral instrument to administer, and how to summarize its output. The second choice is rarely examined.

The conventional summary is **severity**. The Western Aphasia Battery¹ yields subscores for fluency, auditory comprehension, repetition, and naming, each collapsing a heterogeneous performance into a single ordinal rating.

An alternative is to summarize the same performance by **error type**: the proportion of responses that are phonologically related, semantically related, neologistic, or absent. Error type is the currency of process-level accounts of aphasia^{3,4}, and both the 10-item WAB sentence-completion and responsive-speech subtests and the 175-item Philadelphia Naming Test² can be scored this way.

These summaries are typically treated as interchangeable descriptions differing in **granularity**, with item count assumed to govern anatomical resolution. We tested that assumption directly.

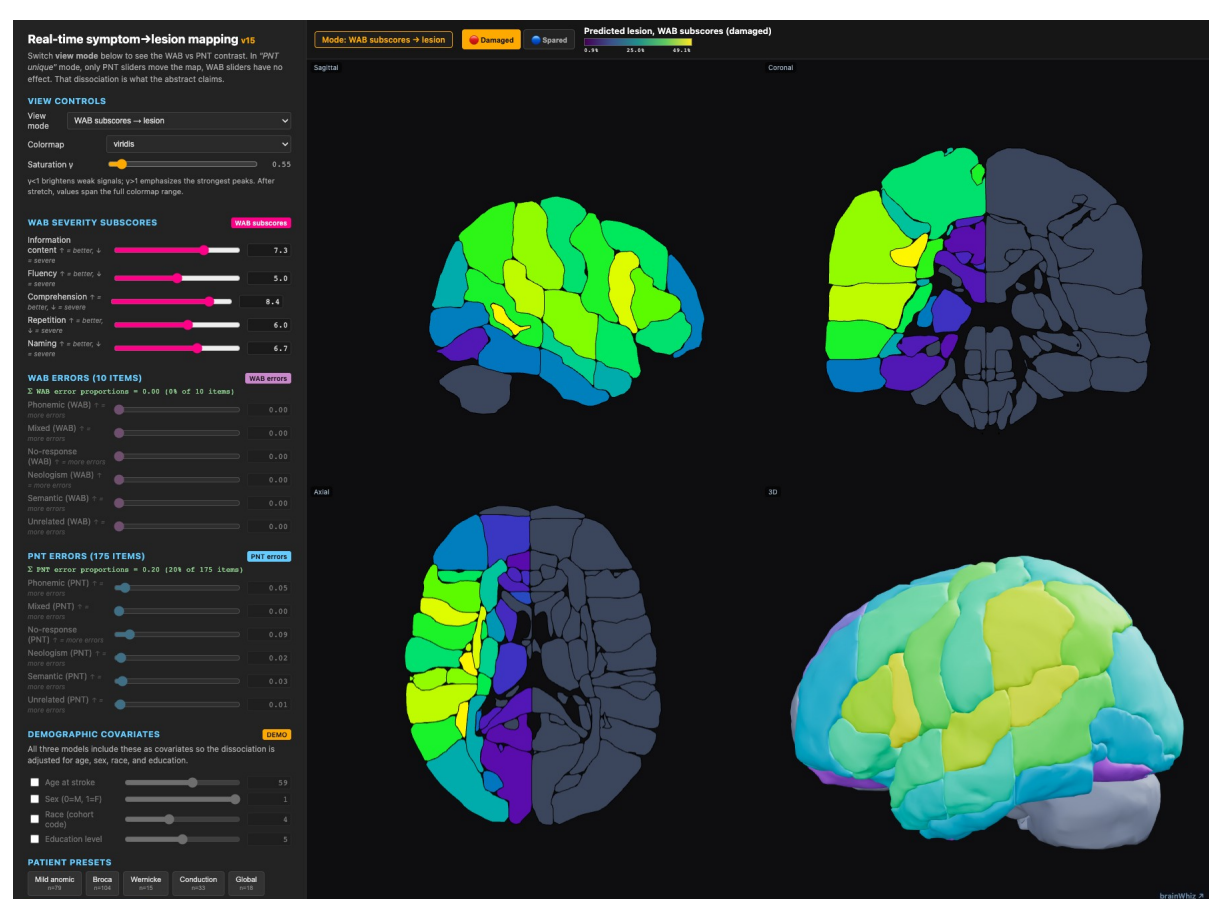
Two questions. Does the choice between severity-aggregated and error-typed behavior change the anatomy you can recover? And does the **number of test items** matter?

PARTICIPANTS

296 chronic post-stroke CSTAR participants, unilateral LH stroke, lesions segmented on the JHU atlas^{12,13} (189 regions; the 87 left-hemisphere non-ventricle ROIs used here).

- WAB severity subscores: **N = 290**
- WAB error proportions: **N = 188**
- PNT error proportions: **N = 213**
- All three lenses present: **N = 133** (paired contrasts)

Live demo at this poster



The model running in the browser at this poster's demo link, rendered in brainWhiz: three planes plus a 3D surface, every left-hemisphere region coloured by predicted lesion load. Move a WAB severity slider and the whole perisylvian territory brightens together; move a PNT error slider and only specific atom territories light up. That dissociation is the claim, live.

Ask me to run it. Presets: Broca, Wernicke, Conduction, Anomic, Global.

METHOD

1. Lesion atoms. Non-negative matrix factorization^{9,10} decomposes each patient's lesion pattern into a recipe of **five spatially coherent atoms**, reproducible across split-halves ($r = 0.83 \pm 0.07$).

2. Anatomical fingerprint. For each behavioral score, one partial Spearman correlation per atom, controlling for the other four atoms **and total lesion volume**: a five-number signature of where that behavior points.

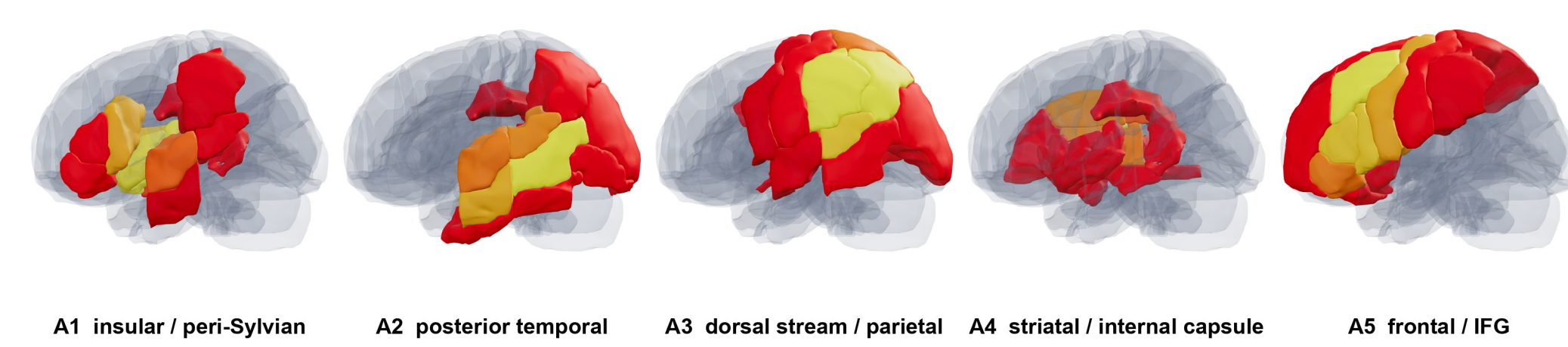
3. Headline statistic. The mean off-diagonal correlation between fingerprints within a lens: how anatomically **indistinguishable** that lens's behaviors are from one another. Bootstrap 95% CIs; Benjamini–Hochberg¹¹ correction on individual cells; pipeline in scikit-learn¹⁴.

Atom composition

Atom	Top-loading ROIs
A1 insular / peri-Sylvian	Plns, Ins, EC
A2 posterior temporal	PSMG, MTG, PSTG
A3 dorsal stream	SLF, SMG, AG
A4 striatal / int. capsule	Put, PLIC, GP
A5 frontal / IFG	MFG, IFG-tri, IFG-oper

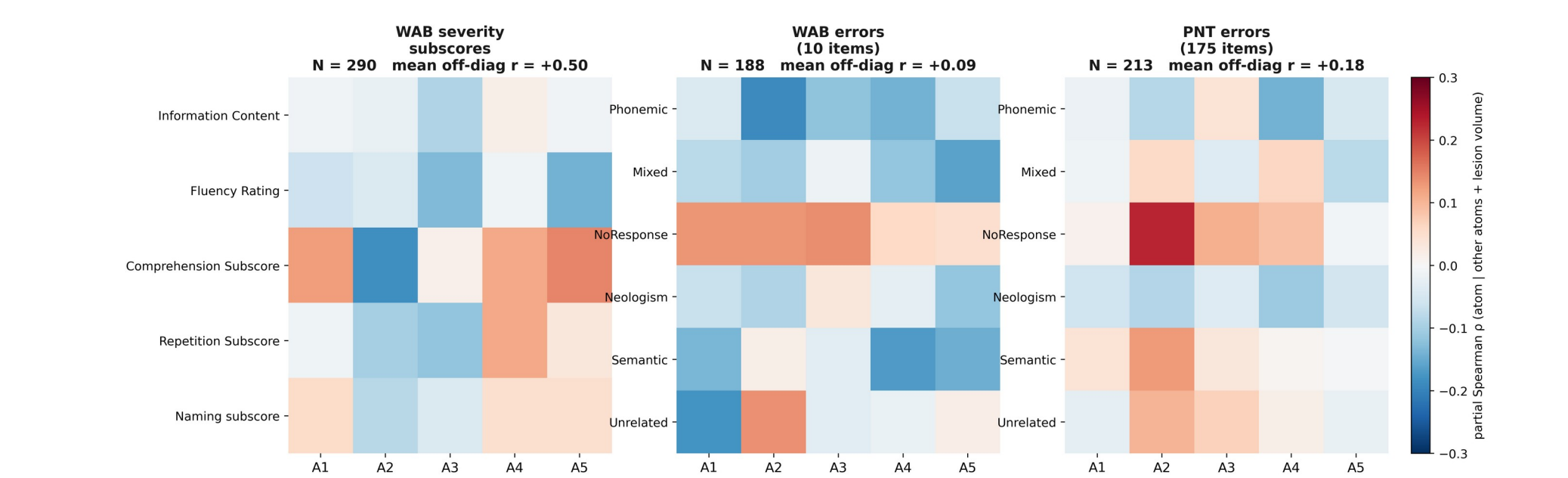
RESULTS

Five lesion atoms



The five lesion atoms from non-negative matrix factorization, each drawn on a glass brain with brainWhiz: colour is the atom's NMF weight on each of the 87 left-hemisphere regions (yellow = highest); uncoloured regions are faded so deep atoms show. Every patient's lesion map is a non-negative mixture of these five footprints: 87 regions become five numbers.

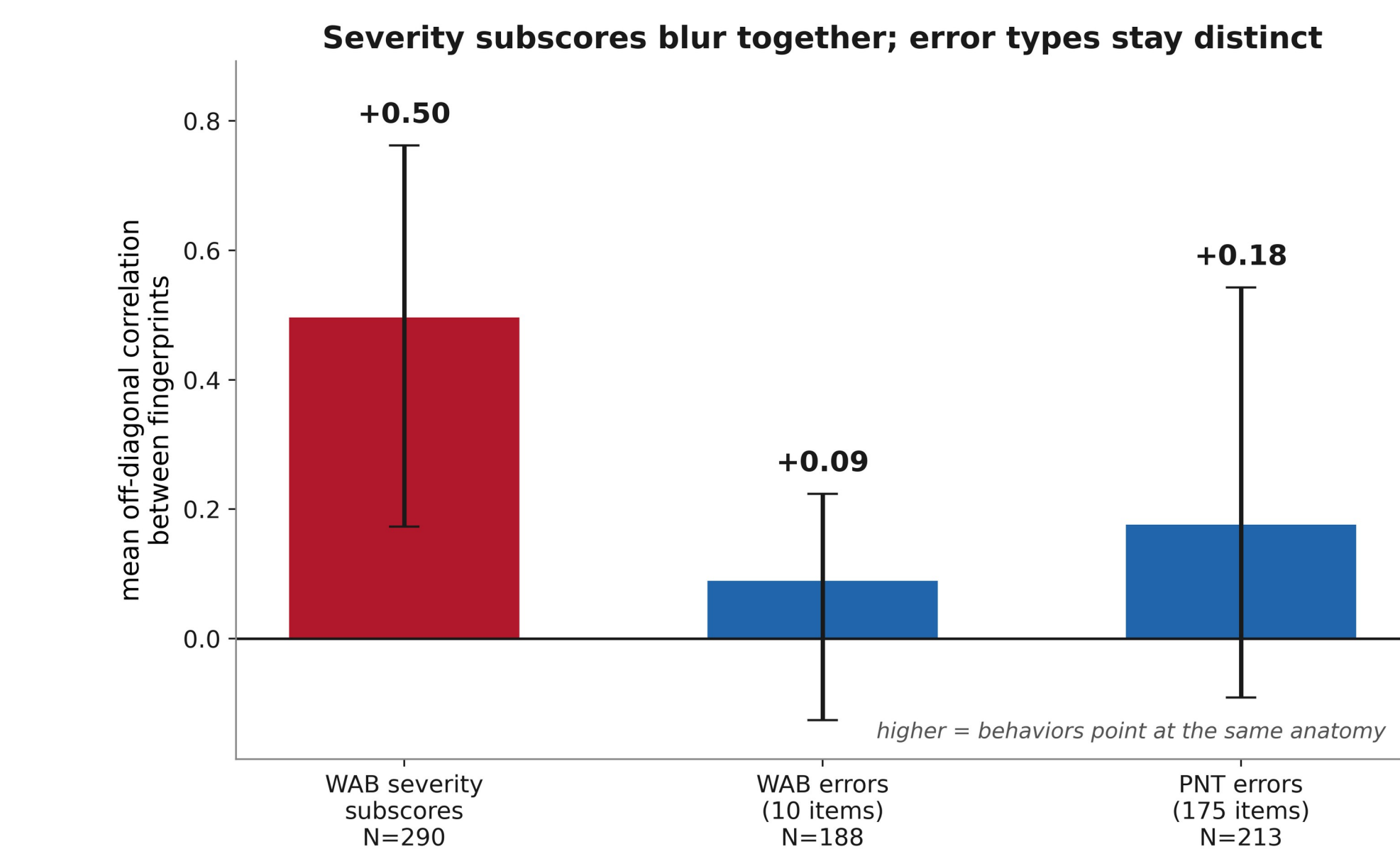
Fingerprints by lens



Each row is one behavioral measure's anatomical fingerprint: its partial Spearman correlation with each of the five atoms, controlling for the other four atoms and total lesion volume. Red = a higher score goes with more damage in that atom, blue = less. Rows that look alike are behaviors pointing at the same anatomy.

The five severity subscores point at **largely the same anatomy**, with small differences in emphasis. Error types point at **different territories**.

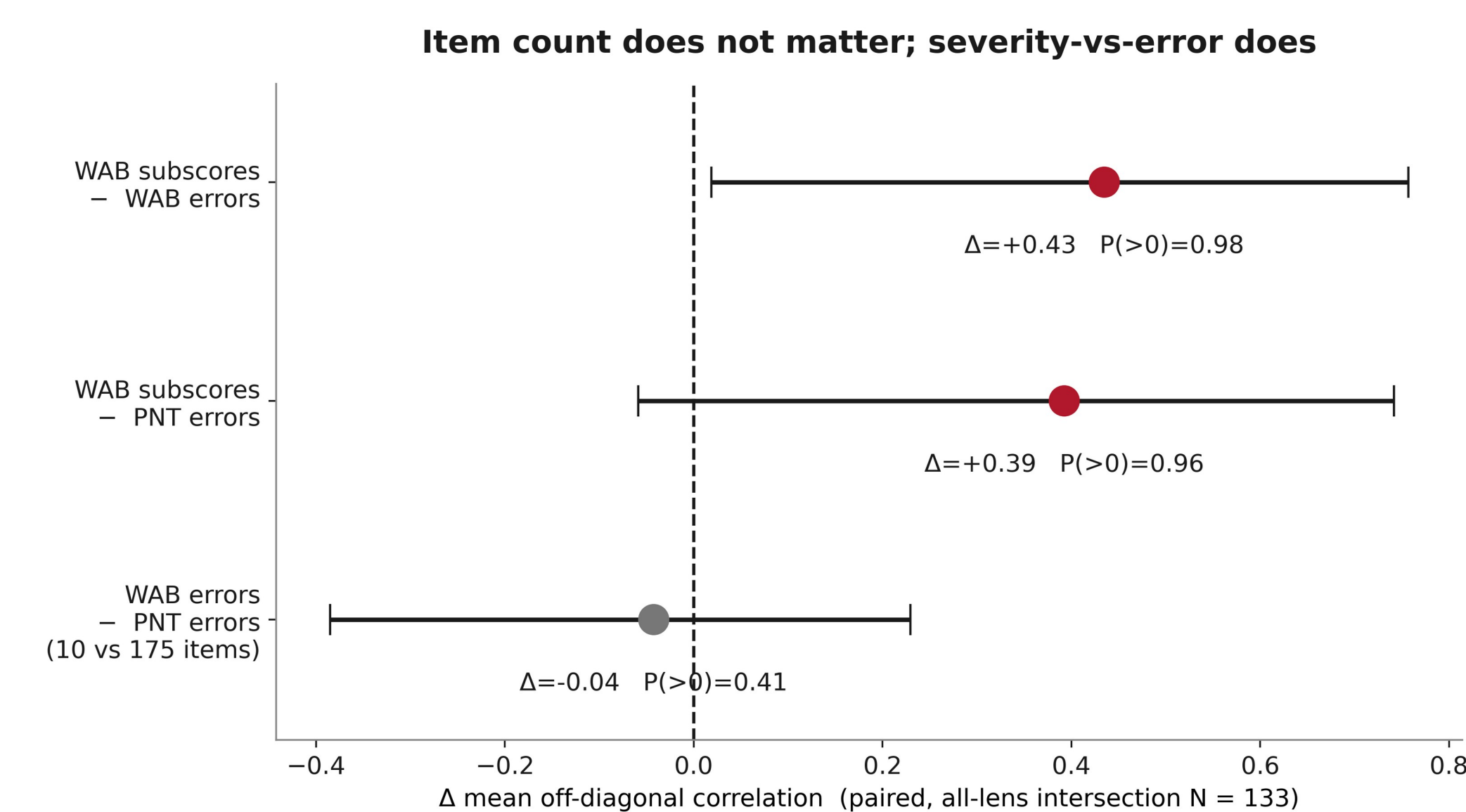
Severity blurs, errors resolve



Mean correlation between the fingerprints of different behaviors within the same lens, with bootstrap 95% CIs over participants. A high value means the behaviors in that lens are anatomically interchangeable; a value near zero means each one points somewhere different.

Item count is not the axis

WAB severity subscore fingerprints were **highly similar to each other** (mean off-diagonal $r = +0.50$, 95% CI [+0.17, +0.76]). WAB error fingerprints were **not** (+0.09 [−0.13, +0.22]), nor were PNT error fingerprints (+0.18 [−0.09, +0.54]).



Paired differences in that same statistic, computed on the 133 participants carrying all three lenses so the comparison is within-subject. Points are the difference, horizontal bars the bootstrap 95% CI, and $P(>0)$ the proportion of bootstrap draws above zero. Positive means the first lens blurs more.

Item count is not the axis. WAB errors and PNT errors did not differ from each other ($\Delta = -0.04$ [−0.39, +0.23], $P(>0) = 0.41$) despite a 17-fold difference in test length.

Individual atom-by-behavior partials were modest ($|p| \leq 0.23$) and almost none survived BH correction: the finding lives at the level of **fingerprint distinctness**, not any single region claim.

What this changes in practice. A study that summarizes behavior as WAB severity is, in effect, asking one anatomical question five times over. The same 296 patients support **fundamentally different anatomical inferences** depending on which behavioral lens is used: a caution for any lesion–behaviour study^{8,15,16}, the component-process approach included^{5,6,7}.

How solid are the atoms?

The five-atom solution is not an artifact of one fit: atom maps matched across 50 independent split-halves at $r = 0.83 \pm 0.07$. Atoms are fit once on all lesion maps and then held fixed while participants are bootstrapped, so a resample cannot permute atom identity underneath the statistic.

Every number on this poster

Lens	N	mean r	95% CI
WAB severity subscores	290	+0.50	+0.17 to +0.76
WAB errors (10 items)	188	+0.09	−0.13 to +0.22
PNT errors (175 items)	213	+0.18	−0.09 to +0.54

DISCUSSION

What you measure decides what anatomy you can see, and the dimension that matters is **not the test or its item count**. It is whether behavior is summarized as **severity** or as **error type**.

Severity subscores blur into a common general-impairment signal. Error proportions from a 10-item subtest and from a 175-item test preserve distinct anatomical signals at comparable resolution.

The result lives at the level of **fingerprint distinctness across behaviors**, not any single brain-region claim: individual partial correlations were modest and almost none survived correction.

FUTURE DIRECTIONS

- Error-type scoring of further batteries**, to test whether the effect generalizes beyond naming and repetition
- Finer error taxonomies**, to ask how much anatomical resolution error type can actually buy
- Replication** in an independent cohort carrying all three lenses
- A prospective test: does error-typed phenotyping improve inverse lesion prediction over severity scores?

Severity scores remain useful clinical anchors. But when the goal is to relate deficits to specific anatomy, **error-typed behavior, even from a 10-item subtest, should be the default**.

Limitations

Each fingerprint correlation is computed over only five atoms, which is why the intervals are wide: read the contrasts, not the individual values. Only **2 of 85** atom-by-behavior partials survive Benjamini–Hochberg correction, so this is a claim about whole fingerprints, not any one region.

Atoms summarise 87 regions in five components, so finer anatomy is invisible here, and all 296 participants come from a single site.

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This poster's demo



brainWhiz viewer



Lab website